A Mini-Project report on

FLOOD PREDICTION IN KERALA

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IN
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This is to certify that the Mini Project report **"FLOOD PREDICTION IN KERALA"** being submitted by **G. Sampath (19P61A0474), G. Samanvai (19P61A0472), J. Suma Saraswathi (19P61A0488)** in partial fulfilment for the award of the Degree of Bachelor of Technology in **ELECTRONICS & COMMUNICATION ENGINEERING** to Jawaharlal Nehru Technological University is a record of a bonafide work carried out by them under my guidance and supervision. The result embodied in this project report has not been submitted to any other University/Institution for the award of any Degree/Diploma.

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DECLARATION

We hereby declare that this Mini Project Report titled **"FLOOD PREDICTION IN KERALA"** submitted by us to the Department of **Electronics and Communication Engineering**, VBIT, Aushapur, Under JNTUH, is a Bonafide work undertaken by and it is not submitted to any other University or institution for the award of any degree or diploma.

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ABSTRACT

Floods are among the most destructive natural disasters, which are highly complex to model. The research on the advancement of flood prediction models contributed to risk reduction, policy suggestion, minimization of the loss of human life, and reduction of the property damage associated with floods. To mimic the complex mathematical expressions of physical processes of floods, during the past two decades, machine learning (ML) methods contributed highly in the advancement of prediction systems providing better performance and cost-effective solutions. Due to the vast benefits and potential of ML, its popularity dramatically increased among hydrologists. Researchers through introducing novel ML methods and hybridizing of the existing ones aim at discovering more accurate and efficient prediction models. The main contribution of this paper is to demonstrate the state of the art of ML models in flood prediction and to give insight into the most suitable models. In this paper, the literature where ML models were benchmarked through a qualitative analysis of robustness, accuracy, effectiveness, and speed are particularly investigated to provide an extensive overview on the various ML algorithms used in the field. The performance comparison of ML models presents an in-depth understanding of the different techniques within the framework of a comprehensive evaluation and discussion. As a result, this paper introduces the most promising prediction methods for both long-term and short-term floods. Furthermore, the major trends in improving the quality of the flood prediction models are investigated. Among them, hybridization, data decomposition, algorithm ensemble, and model optimization are reported as the most effective strategies for the improvement of ML methods. This survey can be used as a guideline for hydrologists as well as climate scientists in choosing the proper ML method according to the prediction task.

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1.1 Introduction to Flood Prediction

Flood predictions require several types of data; The amount of rainfall occurring on a real-time basis. The rate of change in river stage on a real-time basis, which can help indicate the severity and immediacy of the threat. Knowledge about the type of storm producing the moisture, such as duration, intensity and areal extent, which can be valuable for determining possible severity of the flooding. Knowledge about the characteristics of a river's drainage basin, such as soil-moisture conditions, ground temperature, snowpack, topography, vegetation cover, and impermeable land area, which can help to predict how extensive and damaging a flood might become. Flood forecasting is the use of forecasted precipitation and streamflow data in rainfall-runoff and streamflow routing models to forecast flow rates and water levels for periods ranging from a few hours to days ahead, depending on the size of the watershed or river basin.[1] Flood forecasting can also make use of forecasts of precipitation in an attempt to extend the lead-time available. Flood forecasting is an important component of flood warning, where the distinction between the two is that the outcome of flood forecasting is a set of forecast time-profiles of channel flows or river levels at various locations, while "flood warning" is the task of making use of these forecasts to tell decisions on warnings of floods.

Real-time flood forecasting at regional area can be done within seconds by using the technology of artificial neural network.Effective real-time flood forecasting models could be useful for early warning and disaster prevention.

IA can assist us in predicting floods. A case study of the Red River of the north applied Deep Learning algorithms to predict flood-water-level using previous data from Pembina, Drayton, and Grand Forks.

The challenges in using SSMPs in riverine flood prediction have been already summarized in Massari et al. (2015) and Li et al. (2016). Particularly, we believe that the full characterization of the spatial/temporal variability of the error in SSMPs can benefit their use in rainfall-runoff modeling. Several previous studies considered a constant SSMP error in time and/or space (e.g., Brocca et al., 2012), which could be a difficult assumption to relax. Similarly, it is difficult to quantify the modeling error. For example, in an Ensemble Kalman Filter (EnKF, Evensen, 2009) approach, the model error is usually obtained by randomly perturbing the model forcing data, states, and/or parameters. However, the entity, type, and distribution of those perturbations are often subjective.

A recent example of SSMP assimilation, obtained from ASCAT for flood modeling in the Mediterranean region has been published by Loizu et al. (2018), where two different rainfall-runoff models (MISDc and TOPLATS, Brocca et al., 2011; Peters-Lidard et al., 1997) have been applied to two catchments in Spain and Italy. Fig. 10.3 shows the summary of the results for the two catchments and models in terms of Nash–Sutcliffe Efficiency (NSE) as a function of the assumed observation error in the satellite observations. When using MISDc, a slight improvement can be obtained in the Spanish catchment, whereas the model performance in the Italian catchment is very good to start with and thus difficult to improve. The TOPLATS performance is generally poorer and benefits from data assimilation (i.e., in the Italian catchment NSE increases from 0.71 to 0.85 after the assimilation of SSMP). These results, together with those highlighted by Massari et al. (2015) and Li et al. (2016), clearly underline the need of continuing and advancing research activities on this topic to provide guidelines for the optimal use of SSMPs in a data assimilation framework for flood modeling

Floods are a widespread natural hazard that has a devastating socioeconomic toll worldwide. Flood prediction modeling and monitoring are powerful tools for mitigating flood damage. The ongoing development of ML methods over the past decades, along with the availability of long-term hydrological parameters such as precipitation and streamflow, has shown a high level of ability to predict floods.10,19,20 Indeed, comparing the performance of physical and traditional statistical models with ML prediction models to flood prediction shows that the ML models have greater accuracy.21 In this section, an overview of some common ML algorithms, including LSTM, RF, SVM, and CNN, for flood prediction and monitoring are presented.

A considerable amount of literature has been published on the use of ML algorithms to predict flood and streamflow magnitudes at different temporal scales, including hourly,22–24 daily,14,25,26 and monthly.27–29 For instance, an LSTM model was chosen to predict the daily streamflow over 531 basins in the United States. They compared their results with the output of the Sacramento Soil Moisture Accounting (SAC-SMA) physical model and found that LSTM improved prediction.19 This is consistent with what has been the findings of another study.20 Furthermore, to improve the performance of the ML modeling procedure, some researchers have used a hybrid method that combines ML algorithms.10,30 Ni et al.31 used a coupled LSTM with CNN and wavelet-LSTM to forecast monthly streamflow. Their results demonstrated that both coupled models improved the accuracy of a single LSTM. In addition, the hybridization of the ML method with the physical model can improve streamflow prediction performance. Yang et al.32 used the hybrid physical-ML (LSTM) method to predict daily discharge. They used the daily discharge output of the physical model as an input to the LSTM model. The results are better when a single LSTM is used.

In the literature, there are examples of applying ML and DL to detect and forecast flood water levels. Because the data being used is an image, most of these studies applied the CNN method. Moy de Vitry et al.33 applied a deep convolutional neural network (DCNN) model to identify floods from surveillance camera systems in urban areas. Their results showed that flood water was well captured, with an 85% correlation. Wang et al.34 forecasted the water level for a lead time of 1 to 6 h using a

dilated causal convolutional neural network (DCCNN). They compared their model with multilayer perceptron (MLP) and SVM models, and their results showed the priority of DCCNN.

Flood inundation and susceptibility maps are necessary for mitigating, planning, and managing flood incidents.35 ML methods with higher performance than physical and numerical methods have been used to map flood susceptibility.10 El-Haddad et al.36 used four different ML methods, including the functional data analysis (FDA), boosted regression tree (BRT), multivariate discriminant analysis (MDA), and general linear model (GLM) to produce flood-prone zones in Egypt, where the FDA shows better performance than other methods. Moghaddam et al.37 produced a flood susceptibility map for the Tashan watershed in southwestern Iran using six different advanced ML methods. The classification and regression tree (CART) has the best skill among RF, generalized additive model (GAM), multivariate adaptive regression splines (MARS), and maximum entropy (ME). Vafakhah et al.38 and Pourghasemi et al.39 applied three (FR, adaptive network-based fuzzy inference system (ANFIS, and RF) and two (BRT and GLM) algorithms to generate flood susceptibility maps, respectively. Rahmati and Pourghasemi40 used a bivariate statistical analysis model, the evidential belief function (EBF), and two ML algorithms, RF and BRT, to produce flood maps. Their results showed that EBFs exhibited better performance. In addition, other methods such as ANN and ANIFIS have been used in different studies to map flood-prone zones.41,42

A few studies have used optimization algorithms to improve the performance of ML techniques. Termeh et al.42 used an ensemble of different optimization methods, such as genetic algorithm, particle swarm optimization (PSO), and ant colony optimization with ANFIS. In this study, an ensemble of ANFIS-PSO was the best model. Yuan et al.43 forecasted monthly runoff using LSTM and the ant lion model (ALO), which is an optimization algorithm. Their results demonstrated that LSTM-ALO performed better than LSTM.

1.2 Aim

Aim of the project is to predict floods using previous data sets.

1.3 Objectives:

The objectives of the project are illustrated below:

- Collecting the data from GitHub
- Import numpy ,matplot ,keras libraries
- Train the data using SVM
- Save the model for future Implementation
- Testing the accuracy of the model
- Prediction of floods

Chapter 2

2.1 Support Vector Machine

In machine learning, support vector machines (SVMs, also support vector networks are supervised learning models with associated learning algorithms that analyze data for classification and regression analysis. Developed at AT&T Bell Laboratories by Vladimir Vapnik with colleagues (Boser et al., 1992, Guyon et al., 1993, Cortes and Vapnik, 1995,[1] Vapnik et al., 1997 SVMs are one of the most robust prediction methods, being based on statistical learning frameworks or VC theory proposed by Vapnik (1982, 1995) and Chervonenkis (1974). Given a set of training examples, each marked as belonging to one of two categories, an SVM training algorithm builds a model that assigns new examples to one category or the other, making it a non-probabilistic binary linear classifier (although methods such as Platt scaling exist

to use SVM in a probabilistic classification setting). SVM maps training examples to points in space so as to maximise the width of the gap between the two categories. New examples are then mapped into that same space and predicted to belong to a category based on which side of the gap they fall.

In addition to performing linear classification, SVMs can efficiently perform a non-linear classification using what is called the kernel trick, implicitly mapping their inputs into high-dimensional feature spaces.

When data are unlabelled, supervised learning is not possible, and an unsupervised learning approach is required, which attempts to find natural clustering of the data to groups, and then map new data to these formed groups. The support vector clustering algorithm, created by Hava Siegelmann and Vladimir Vapnik, applies the statistics of support vectors, developed in the support vector machines algorithm, to categorize unlabeled data.

More formally, a support vector machine constructs a hyperplane or set of hyperplanes in a high or infinite-dimensional space, which can be used for classification, regression, or other tasks like outliers detection. Intuitively, a good separation is achieved by the hyperplane that has the largest distance to the nearest training-data point of any class (so-called functional margin), since in general the larger the margin, the lower the generalization error of the classifier.

Classification of ML Methods and Applications

The most popular ML modeling methods for flood prediction were identified in the previous section, including ANFIS, MLP, WNN, EPS, DT, RF, CART, and ANN. Figure 3 presents the major ML methods used for flood prediction, and the number of corresponding articles in the literature over the last decade. This figure was designed to communicate to the readers which ML methods increased in popularity among hydrologists for flood modeling within the past decade.

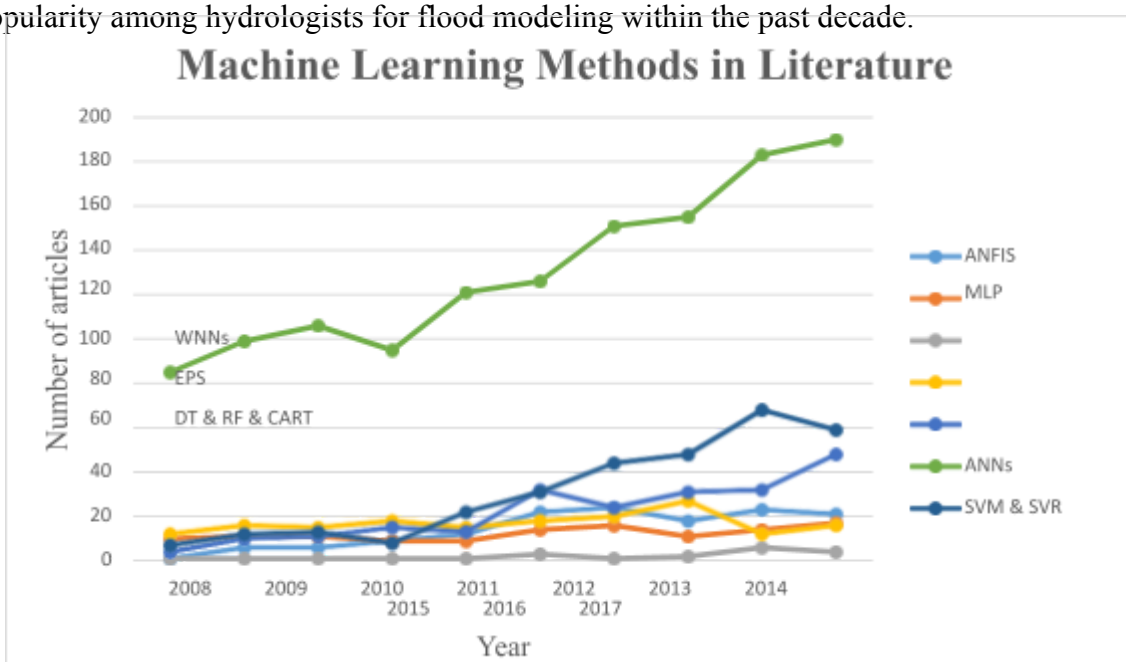


Figure 3. Major ML methods used for flood prediction in the literature. Reference year: 2008 (source:

Scopus).

Considering the ML methods for application to floods, it is apparent that ANNs, SVMs, MLPs, DTs, ANFIS, WNNs, and EPSs are the most popular. These ML methods can be categorized as single and hybrid methods. In addition to the fundamental hybrid ML methods, i.e., ANFIS, WNNs, and basic EPSs, several different research strategies for obtaining better prediction evolved [137]. The strategies involved developing hybrid ML models using soft computing techniques, statistical methods, and physical models rather than individual ML approaches, whereby the extra components complement each other with respect to their drawbacks and shortcomings. The success of such hybrid approaches motivated the research community to explore more advanced hybrid models. Figure 4 presents the progress of single vs. hybrid ML methods for flood prediction in the literature over the past decade. The figure shows an apparent continuous increase and notable progress in using novel hybrid methods. Through Figure 4, the taxonomy of our research was justified, based on distinguishing hybrid and single ML prediction models.

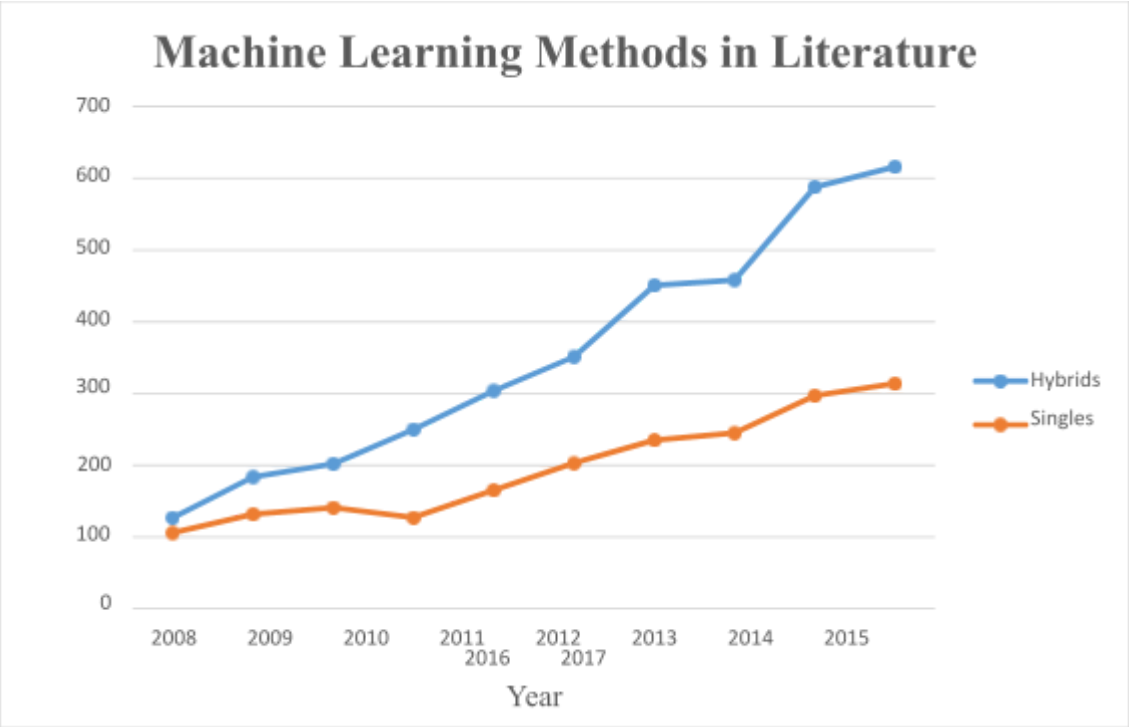


Figure 4. The progress of single vs. hybrid ML methods for flood prediction in the literature. Reference year: 2008 (source: Scopus).

Furthermore, the types of prediction are often studied with different lead-time predictions due to the flood. Real-time, hourly, daily, weekly, monthly, seasonal, annual, short-term, and long-term are the terms most often used in the literature. Real-time prediction is concerned with anywhere between few minutes and an hour preceding the flood. Hourly predictions can be 1–3 h ahead of the flood forecasting lead time or, in some cases, 18 h or 24 h. Daily predictions can be 1–6 days ahead of the

forecast. Monthly forecasts can be, for instance, up to three months. In hydrology, the definitions of short-term and long-term in studying the different phenomena vary. Short-term predictions for floods often refer to hourly, daily, and weekly predictions, and they are used as warning systems. On the other hand, long-term predictions are mostly used for policy analysis purposes. Furthermore, if the prediction leading time to flood is three days longer than the confluence time, the prediction is considered to be long-term [37,58]. From this perspective, in this study, we considered a lead time greater than a week as a long-term prediction. It was observed that the characteristics of the ML methods used varied significantly according to the period of prediction. Thus, dividing the survey on the basis of short-term and long-term was essential.

Here, it is also worth emphasizing that, in this paper, the prediction lead-time was classified as "short-term" or "long-term". Although flash floods happen in a short period of time with great destructive power, they can be predicted with either "short-term" or "long-term" lead times to the actual flood. In fact, this paper is concerned with the lead times instead of the duration or type of flood. If the lead-time prediction to a flash flood was short-term, then it was studied as a short-term lead time. However, sometimes flash floods can be predicted with long lead times. In other words, flash floods might be predicted one month ahead. In this case, the prediction was considered as long-term. Regardless of the type of flood, we only focused on the lead time.
In this study, the ML methods were reviewed using two classes—single methods and hybrid methods. Figures 5 and 6 represent the taxonomy of the research.

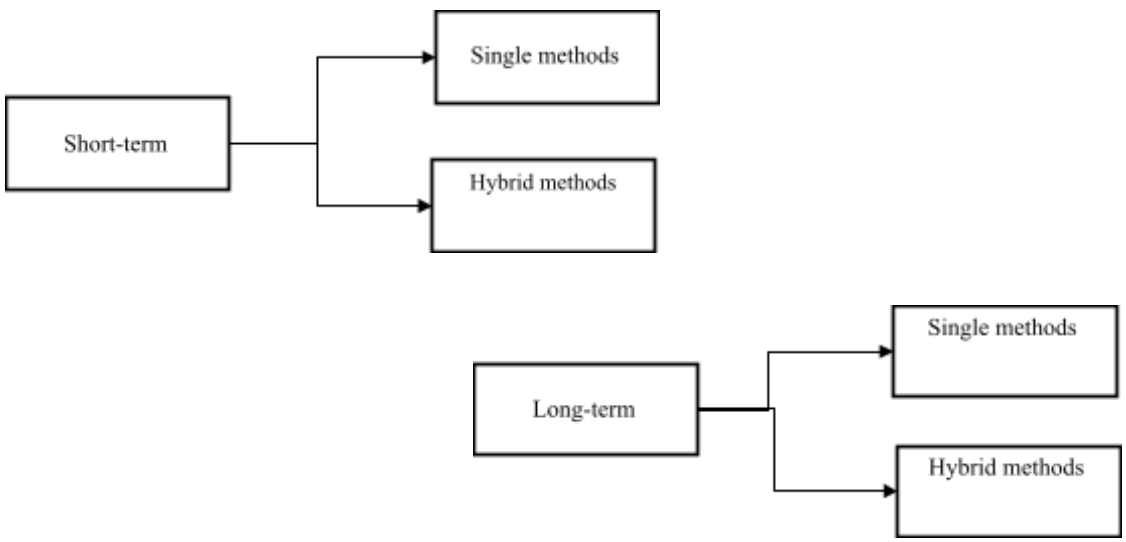


Figure 5. Taxonomy of the survey—ML methods for flood prediction.

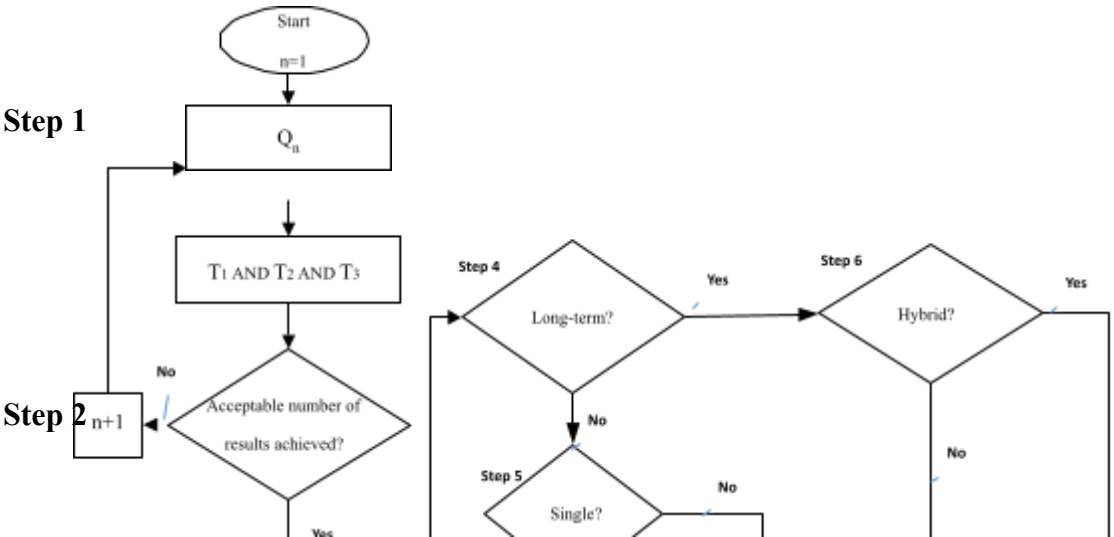


Figure 6. Taxonomy of the survey.

Step 1 involved running the queries one by one; step 2 involved checking the results of the search, and initiating the next search; step 3 involved identifying the comparative studies on ML models of prediction, refining the results and building the database; step 4 involved identifying whether it was a long-term or short-term prediction; steps 5 and 6 involved identifying if it was a single or hybrid method, constructing Table 1, and step 7 involved constructing the other Tables. The four tables provide the list of studies on different prediction techniques, which entail the organized comprehensive surveys of the literature.

Short-Term Flood Prediction with ML

Short-term lead-time flood predictions are considered important research challenges, particularly in highly urbanized areas, for timely warnings to residences so to reduce damage [146]. In addition, short-term predictions contribute highly to water recourse management. Even with the recent improvements
in numerical weather prediction (NWP) models, artificial intelligence (AI) methods, and ML, short-term prediction remains a challenging task [147–152]. This section is divided into two subsections—single and hybrid methods of ML—to individually investigate each group of methods.

Short-Term Flood Prediction Using Single ML Methods

To gain insight into the performance of ML methods, a comprehensive comparison was required to investigate ML methods. Table 1 presents a summary of the major ML methods, i.e., ANNs, MLP, nonlinear autoregressive network with exogenous inputs (NARX), M5 model trees, DTs, CART, SVR, and RF, followed by a comprehensive performance comparison of single ML methods in short-term flood prediction. A revision and discussion of these methods follow so as to identify the most suitable methods presented in the literature.

Table 1. Short-term predictions using single machine learning (ML) methods.

Modeling Technique	Reference	Flood Resource Variable	Prediction Type	Region
ANN vs. statistical	[1]	Streamflow and flash food	Hourly	USA
ANN vs. traditional	[44]	Water and surge level	Hourly	Japan
ANN vs. statistical	[149]	Flood	Real-time	UK
ANN vs. statistical	[150]	Extreme flow	Hourly	Greece

FFANN vs. ANN	[151]	Water level	Hourly	India
ANN vs. T–S	[4]	Flood	Hourly	India
ANN vs. AR	[153]	Stage level and streamflow	Hourly	Brazil
MLP vs. Kohonen NN	[154]	Flood frequency analysis	Long-term	China
BPANN	[155]	Peak flow of flood	Daily	Canada
BPANN vs. DBPANN	[156]	Rainfall–runoff	Monthly and daily	China
BPANN	[157]	Flash flood	Real-time	Hawaii
BPANN	[158]	Runoff	Daily	India
ELM vs. SVM	[159]	Streamflow	Daily	China
BPANN vs. NARX	[160,161]	Urban flood	Real-time	Taiwan
FFANN vs. Functional	[162]	River flows	Real-time	Ireland
Recurrent NN vs. Z–R relation	[163]	Rainfall prediction	Real-time	Taiwan
ANN vs. M5 model tree	[164]	Peak flow	Hourly	India
NBT vs. DT vs.	[165]	Flash flood	Real-time, hourly	Austria
DTs vs. NBT vs. ADT vs.	[166]	Flood	Hourly/daily	Iran
		River flow and rainfall–runoff		
MLP vs. MLR	[98]	River runoff	Hourly	Morocco
MLP vs. WT vs. MLR vs.	[169]	River flood forecasting	Daily	Canada
ANN vs. MLP	[170]	River level	Hourly	Ireland
MLP vs. DT vs. CART	[171]	Flood during typhoon	Rainfall–runoff	China
SVM vs. ANN	[120]	Rainfall extreme events	Daily	India
ANN vs. SVR	[48]	Flood	Daily	Canada
RF vs. SVM	[69]	Rainfall	Hourly	Taiwan

Kim and Barros [148] modified an ANN model to improve flood forecasting short-term lead time through consideration of atmospheric conditions. They used satellite data from the ISCCP-B3 dataset [172]. This dataset includes hourly rainfall from 160 rain gauges within the region. The ANN was reported to be considerably more accurate than the statistical models. In another similar work, Reference [44] developed an ANN forecast model for hourly lead time. In their study, various datasets were used, consisting of meteorological and hydrodynamic parameters of three typhoons. Testing of the ANN forecast models showed promising results for 5-h lead time. In another attempt, Danso-Amoako [1] provided a rapid system for predicting floods with an ANN. They provided a reliable forecasting tool for rapidly assessing floods. An R^2 value of 0.70 for the ANN model proved that the tool was suitable for predicting flood variables with a high generalization ability. The results of [149] provides similar conclusions. Furthermore, Panda, Pramanik, and Bala [151] compared the accuracy of ANN with FFANN, and the results were benchmarked with the physical model of MIKE 11 for short-term water level prediction. This dataset includes the hourly discharge and water level between 2006 and 2009. The data of the year 2006 was used for testing root-mean-square error (RMSE). The results indicated that the FFANN performed faster and relatively more accurately than the ANN model. Here, it is worth mentioning that the overall results indicated that the neural networks were superior compared to the one-dimensional model MIKE 11. Nevertheless, there were great advancements reported in the implementation of two-dimensional MIKE 11 [8].

Kourgialas, Dokou, and Karatzas [150] created a modeling system for the prediction of extreme flow

based on ANNs 3 h, 12 h, and 19 h ahead of the flood. They analyzed five years of hourly data to investigate the ANN effectiveness in modeling extreme flood events. The results indicated it to be highly effective compared to conventional hydrological models. Lohani, Goel, and Bhatia [4] improved the real-time forecasting of rainfall–runoff of foods, and the results were compared to the T–S fuzzy model and the subtractive-clustering-based T–S (TSC-T–S) fuzzy model. They, however, concluded that the fuzzy model provided more accurate predictions with longer lead time. The hourly rainfall data from 1989 to 1995 of a gauge site, in addition to the rainfall during a monsoon, was used. Pereira Filho and dos Santos [153] compared the AR model with an ANN in simulating forecast stage level and streamflow. The dataset was created from independent flood events, radar-derived rainfall, and streamflow rain gauges available between 1991 and 1995. The AR and ANN were employed to model short-term flood in an urban area utilizing streamflow and weather data. They showed that the ANN performed better in its verification and it was proposed as a better alternative to the AR model. Ahmad and Simonovic [155] used a BPNN for predicting peak flow utilizing causal meteorological parameters. This dataset included daily discharge data for 1958–1997 from gauging stations. BPNN proved to be a fast and accurate approach with the ability of generalization for application to other locations with similar rivers. Furthermore, to improve the simulation of daily streamflow using BPNN, Reference [156] used division-based backpropagation to obtain satisfying results. The raw data of local evaporation and rainfall gauges of six years were used for the short-term flood prediction of a streamflow time series. The dataset of one decade from 1988 was used for training and the dataset of five subsequent years was used for testing. The BPNN model provided promising results; however, it lacked efficiency in using raw data for the time-series prediction of streamflow. In addition, Reference [157] showed the application of BPNN for assessing flash floods using measured data. This dataset included 5-min-frequency water quality data and 15-min-frequency rainfall data of 20 years from two rain gauge stations. Their experiments introduced ANN models as simple ML methods to apply, while simultaneously requiring expert knowledge by the user. In addition, their ANN prediction model showed great ability to deal with a noisy dataset. Ghose [158] predicted the daily runoff using a BPNN prediction model. The data of daily water level of two years from 2013–2015 were used. The accurate BPNN model was reported with an efficiency of 96.4% and an R^2 of 0.94 for flood prediction.

Pan, Cheng, and Cai [159] compared the performances of ELM and SVM for short-term streamflow prediction. Both methods demonstrated a similar level of accuracy. However, ELM was suggested as a faster method for parameter selection and learning loops. Reference [154] also conducted a comparison between fuzzy c-means, ANN, and MLP using a common dataset of sites to investigate ML method efficiency and accuracy. The MLP and ANN methods were proposed as the best methods. Chang, Chen, Lu, Huang, and Chang [160] and Reference [161] modeled multi-step urban flood forecasts using BPNN and a nonlinear autoregressive network with exogenous input (NARX) for hourly forecasts. The results demonstrated that NARX worked better in short-term lead-time prediction compared to BPNN. The NARX network produced an average R^2 value of 0.7. This study suggested that the NARX model was effective in urban flood prediction. Furthermore, Valipour et al. [24] showed how the accuracy of ANN models could be increased through integration with autoregressive (AR) models.

2.3 Applications

SVMs can be used to solve various real-world problems:

- SVMs are helpful in text and hypertext categorization, as their application can significantly reduce the need for labeled training instances in both the standard inductive and transductive settings.[7] Some methods for shallow semantic parsing are based on support vector machines.

- Classification of images can also be performed using SVMs. Experimental results show that SVMs achieve significantly higher search accuracy than traditional query refinement schemes after just three to four rounds of relevance feedback. This is also true for image segmentation systems, including those using a modified version SVM that uses the privileged approach as suggested by Vapnik.Classification of satellite data like SAR data using supervised SVM.

- Hand-written characters can be recognized using SVM.

- The SVM algorithm has been widely applied in the biological and other sciences. They have been used to classify proteins with up to 90% of the compounds classified correctly. Permutation tests based on SVM weights have been suggested as a mechanism for interpretation of SVM models.

- Support vector machine weights have also been used to interpret SVM models in the past. Posthoc interpretation of support vector machine models in order to identify features used by the model to make predictions is a relatively new area of research with special significance in the biological sciences.

The original SVM algorithm was invented by Vladimir N. Vapnik and Alexey Ya. Chervonenkis in 1963. In 1992, Bernhard Boser, Isabelle Guyon and Vladimir Vapnik suggested a way to create nonlinear classifiers by applying the kernel trick to maximum-margin hyperplanes.The "soft margin" incarnation, as is commonly used in software packages, was proposed by Corinna Cortes and Vapnik in 1993 and published in 1995.

Chapter 3

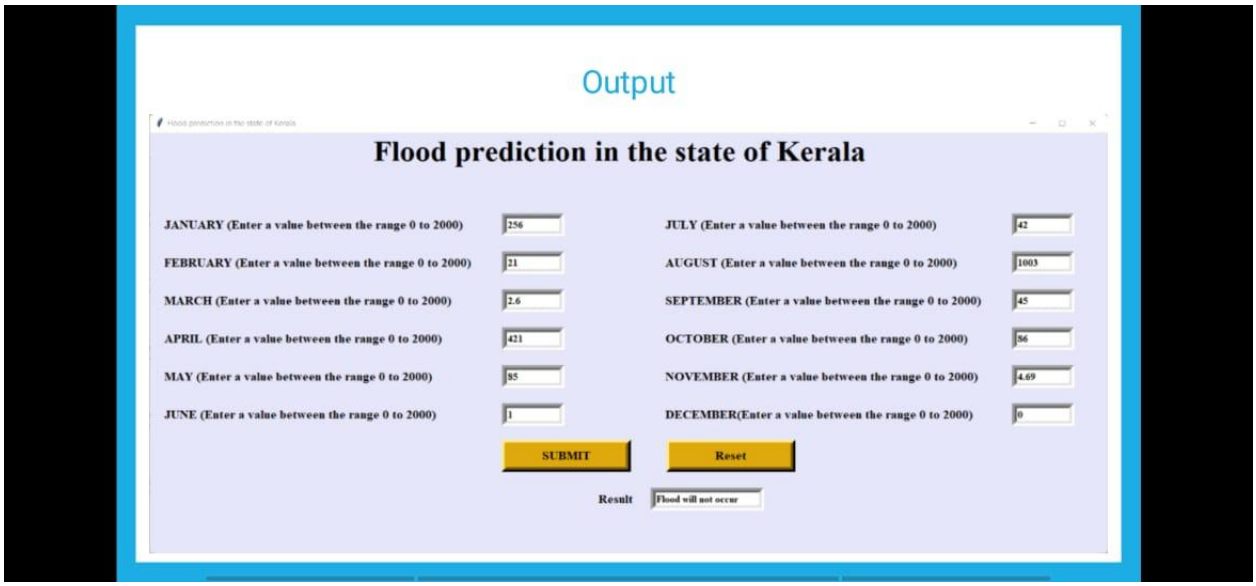
3.1 Training and Testing

```
import numpy as np
import pandas as pd
import pickle
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import
accuracy_score,recall_score,roc_auc_score,confusion_matrix

import seaborn as sns
axis = sns.barplot(x = 'Name', y = 'Score', data =tr_split )
axis.set(xlabel='Classifier Models', ylabel='Accuracy of the Model')
for p in axis.patches:
   height = p.get_height()
   axis.text(p.get_x() + p.get_width()/2, height + 0.01,
'{:1.4f}'.format(height), ha="center")
svm=SVC()
svm.fit(x_train,y_train)
y_predict=svm.predict(x_test)
y_predict
models = []
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier

models.append(('LR', LogisticRegression()))
models.append(('SVC', SVC()))

names = []
scores = []
for name, model in models:
   model.fit(x_train, y_train)
   y_pred = model.predict(x_test)
   scores.append(accuracy_score(y_test, y_pred))
   names.append(name)
tr_split = pd.DataFrame({'Name': names, 'Score': scores})
```

tr_split

Chapter 4

4.1 Results



4.2 Conclusion

SVM has better prediction accuracy and therefore floods can be predicted more accurately. With linear regression or any other algorithms the accuracy is low which is not the case of Support Vector machine learning algorithms.

Using the previous data sets we train our model to predict the flood considering attributes like temperature, rainfall, and other weather patterns.

4.3 Future Scope

Flood Forecasting (FF) using SVm forms an important tool in **reducing vulnerabilities and flood risk** and form an important ingredient of the strategy to "live with floods", thereby contributing to national sustainable development.

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